

CoSpeak: Peer Feedback on Voice Stories to Inform Learning Spoken English

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ABSTRACT

English is officially recognized in India as a means of public communication as it plays an important role in academia, government affairs etc. However, only 10.6% of Indians speak some English according to the 2011 Census report. In this project, primary research was conducted in the context of teaching English to class 12 students from regional language medium schools. The context was understood, difficulties faced in learning, aspects of learning and teaching were identified. The topic of speaking English in daily conversations was prioritized. Existing solutions were analyzed. A design brief consisting of goals based on the research was formulated. Ideas were brainstormed and the final direction was consolidated based on the feedback received. CoSpeak, an interactive mobile application using voice recognition and utilizing collaborative learning was designed iteratively. An evaluation with the teacher and students was conducted using a Wizard of Oz type prototype and future possibilities were identified.

CCS CONCEPTS

• **CSS Concepts:** Human-centred computing - Collaborative and social computing - Collaborative and social computing design and evaluation methods - Ethnographic studies;

KEYWORDS

Voice Stories, Speech To Text, Peer Feedback

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1 INTRODUCTION

English is officially recognized in India as a means of public communication as it plays an important role in academia, government affairs, maintaining unity in the country etc. [8]. This has led to an integration of English in daily communication in the Indian context [9]. However, only 10.6% of Indians speak some English according

to the 2011 Census report [6]. Therefore, several NGOs have been working in this area to help impart spoken English language skills to students from Indian regional language-medium schools to help them access higher education opportunities, career opportunities and foreign opportunities.

For this project, we collaborated with an NGO which works at the intersection of education and employability with schools in the Maharashtra state. The NGO conducts a program called Junior Rural Fellowship (JRF) which focuses on teaching class 12 students spoken English language skills and soft skills (communication, teamwork etc.) such that they can get jobs or study further in English medium colleges. The program has been conducted physically by the NGO trainers in government schools but at present, it is being conducted virtually due to the pandemic. Continuing this program virtually would provide the opportunity of scaling it to a greater number of schools and including different interested students such as first year college students, in the program. The online classes are being conducted using the same reference textbook and activities designed for the delivery of the program in the physical context. An opportunity to design a digital solution focusing on the needs of this area was found.

2 BACKGROUND

Existing studies conducted for learning a second/foreign language showed that the students often understood each word separately, but were unable to link them into meaningful sentences used for communication. Focus on good language learners hampered the development of slower language learners in a class [3].

With the growth of Voice-based technologies, a need for carefully designing voice-based interactions as a part of user interfaces has been suggested [1]. Some work has been done in using interfaces using voice for educational purposes. Studies showed the application of Voice User Interfaces to support creative social play with preschool children of three to four years of age [12].

3 DATA COLLECTION

Data was collected through primary research to get a deeper understanding of the context and identify areas for design interventions. A mixed-method approach was employed to collect data. The study was done in the context of the JRF program. The JRF English classes were conducted online twice a week after school for 1.5 hours for a selected group of ten, class 12 students throughout their final semester for a duration of approximately six months. The study involved participant observation for ten sessions of the English classes followed by semi-structured interviews and questionnaires for the participants. The textbook used for the classes was analyzed. Gaps in the existing digital solutions were identified.

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3.1 Primary Research

The primary research helped to get an understanding of the users and the aspects of learning and teaching used in the context. The students had studied English previously in school as a second/third language and could read and write the script but could not speak English. The motivation of the students was to learn English to communicate in their personal and professional life. The keys aspects of teaching in the JRF classes included repurposing activities from the textbook to make them participatory, contextualizing activities for the Indian context and usage of creativity in the activities to increase engagement. Making mistakes while speaking and guidance to help speak were key aspects of developing the confidence and learning to speak. Collaborative learning was observed in the classes and the students felt comfortable discussing doubts with other students.

3.2 Secondary Research

Secondary research was conducted by analyzing existing digital products to learn English. The existing digital applications consisted of activities like rearranging words to form a sentence by utilizing translation abilities, conversation through AI chat and multiple-choice questions, word meanings, creating books using text and images and listening to audio stories. The existing digital applications did not focus on the utilization of voice technologies, speech input, activities related to speaking or collaborative activities.

4 DESIGN BRIEF

Based on the research, the final design brief was defined to design a digital application for class 12 students in the JRF program to aid in learning spoken English language skills. The application would be used in addition to the English classes.

Through this application, students would learn English as used in day-to-day communication (using a topic-wise reference syllabus). Students would practice speaking English through the designed activities in a safe space at their own pace. Students would be able to utilize the theoretical knowledge in practical situations involving conversation in English. The designed tool would enable collaborative learning.

5 DESIGN PROCESS

Ideas were brainstormed for the application based on the insights from the data collected. Three directions – Creating Audio Stories, Newspaper Reading & Discussion and Unusual Situational Conversation were detailed and presented to the students, teachers and experts for feedback. A participatory process was used to iterate the ideas to arrive at a final direction. The direction of “Creating Audio Stories through Dialogue” was finalized by combining aspects from the initial three directions and building on it.

6 COSPEAK: COLLABORATIVE AUDIO STORIES

CoSpeak, an interactive mobile application was designed. The application enables students learning English to practice speaking by employing a creative storytelling activity in collaboration with other English learners. This helps the student practice speaking

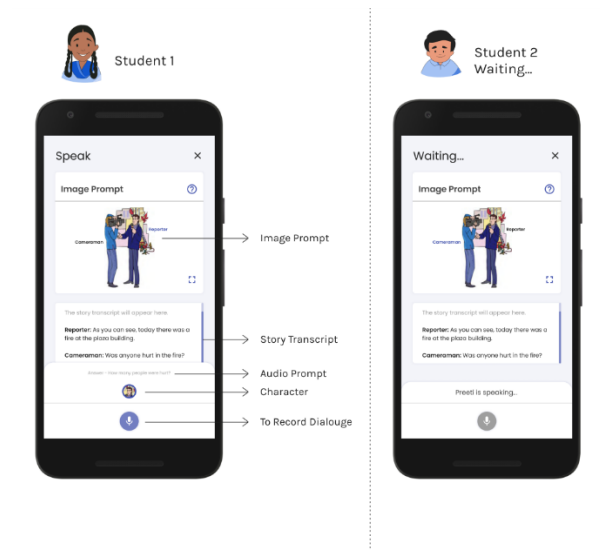


Figure 1: Create a story

English used in daily conversations. The students are also given the responsibility to review their peer’s audio stories leading to collaborative learning and the growth of the community of English learners. Three novel activities were designed as a part of CoSpeak – Create a story, review a story and view a reviewed story.

The application would use speech recognition for inputting every sentence spoken by the student while creating the audio story. Speech to text would be used to create and represent the transcript of the audio stories. The same voice technology would be used for the supporting activities of reviewing an audio story and viewing a reviewed audio story.

6.1 Create a story

Creating a story is an online collaborative activity through which a student can co-create an audio story with a peer. Students are paired together in class by the teacher to create a story. Image prompts consisting of different characters in a situation are provided based on the ongoing topic in class. Each student is assigned a character. To aid the student to start speaking, the beginning of the audio story and an audio prompt to continue the story is provided. The story is created in a turn-by-turn manner as one student speaks out a dialogue while the second student waits (see Fig 1). A help option consisting of a word bank and a word translation feature are also designed.

This method of creating an audio story encourages students to speak at their own pace in a safe environment, utilizes their creativity while maintaining the uncertainty of a real-life conversation scenario.

6.2 Review a story

The review activity is assigned by the teacher to a student in a senior batch. The senior student is given the responsibility of reviewing an audio story created by the paired students. Review a story is a double-blind peer review. The senior student listens to

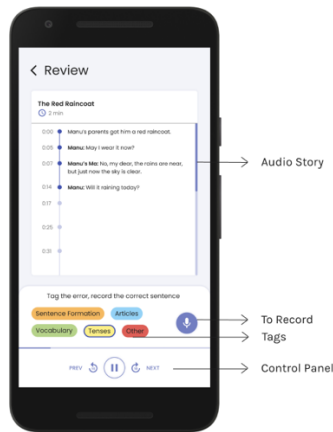


Figure 2: Review a story

the audio story and adds audio comments on identifying mistakes and suggesting improvements (see Fig 2). The audio comments are tagged based on category system of common mistakes.

This method of peer review distributes the responsibility of re-viewing within the student community and ensures regular practice of spoken English skills by senior students as well.

6.3 View a reviewed story

Once the senior student finishes the review, the comments are sent to the paired students. Viewing a reviewed story helps the students view their mistakes and gives them the opportunity to improve and build on their stories by re-recording dialogues (see Fig 3). Once both the paired students finish checking the review and making the changes, the audio story is added to the library and made available to all the English learners.

The three activities together would create collaborative learning. The “create a story” activity is proposed to be done once a week. The image prompts would be updated weekly by repurposing images from an existing topic-wise English textbook. The student pairs to create a story and the reviewers would be randomly assigned every week.

6.4 Supporting features

Other features of CoSpeak include a library for audio stories by students and experts. A profile section for students with information on their level of practice. Self-learning features such re-recording existing audio stories from the library for practice and listening to audio stories are designed. An achievements section including weekly and long-term achievements is proposed to incentivize the students.

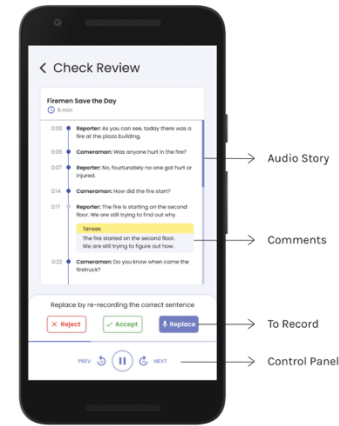


Figure 3: View a reviewed story

7 EVALUATION

For this project, CoSpeak was used in the context of the JRF classes. Unit 4 - Describing events of the JRF textbook and the topic of simple present tense and continuous present tense which was going on in class was used for reference. Two types of image prompts were used – the first image prompt consisting of 2 characters in a defined situation and the second image prompt consisting of multiple scenarios with multiple characters. Evaluation was conducted using a Wizard of type prototype with four students and a teacher.

In the “create a story” activity, students acted out the voice and expressions of the character they spoke for. The students embodied a character assigned at the beginning of the activity and continued with it, turn by turn. In case of multiple characters in an image prompt, choosing a different character to speak for in every turn was confusing for the students. The image prompt with multiple characters in a situation would be more suited for senior students. The students required leading audio prompts to help them speak for the first few turns. The additional help features such as the word bank and the word translation feature were useful according to the students. The students made mistakes while speaking, some of the students immediately realized it and re-spoke the dialogue.

In the “review a story” and “view a reviewed story” activity, students were able to identify their peer’s mistakes in sentences. Apart from grammatical mistakes, they also made suggestions to improve sentences. On receiving the feedback, the paired student utilized the suggestion to build further on it. The activities enabled collaborative learning as the students could learn with each other and from each other. The students were also able to learn from listening to others’ stories during the review activity to set a benchmark and improve their own work.

The teacher and the students appreciated the design of the application and the collaborative nature of the activities. The students

found the ideas to be novel and easily understandable. The teacher and the students found the application helpful to learn spoken English in daily conversations and increase English speaking confidence in a creative and fun way.

8 CONCLUSION AND FUTURE WORK

The contribution of this project to the domain of adults learning English is the design of the three novel activities of creating an audio story through dialogue, peer review of the story and incorporating improvements in the reviewed story for an interactive mobile application using voice recognition. These activities address the topic of speaking English in daily conversations and learning with and through peers which are not addressed or present in the existing digital solutions.

Future work can be done to develop the application utilizing the feedback from the evaluation. CoSpeak can be scaled to create a community of students learning to speak English from all over the country. AI characters can also be developed to help with self-learning for situations where it's difficult to collaborate with a peer. This will be possible with the development of long-term interactions and the turn-taking behavior of voice technologies [1].

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